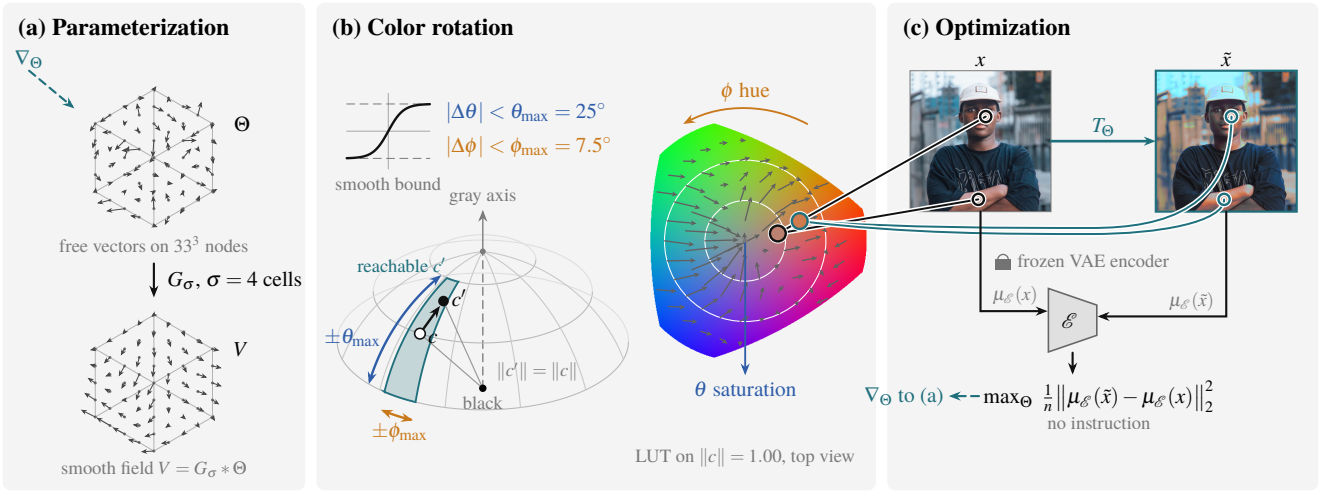

(a) Parameterization
∇_Θ
Θ
free vectors on 33^3 nodes
G_σ, $\sigma = 4$ cells
V
smooth field $V = G_\sigma * \Theta$
(b) Color rotation
$|\Delta\theta| < \theta_{\max} = 25°$
$|\Delta\phi| < \phi_{\max} = 7.5°$
smooth bound
gray axis
reachable c'
$\pm\theta_{\max}$
c'
c
$\|c'\| = \|c\|$
black
$\pm\phi_{\max}$
ϕ hue
θ saturation
LUT on $\|c\| = 1.00$, top view
(c) Optimization
x
$\tilde{x}$
T_Θ
frozen VAE encoder
$\mu_{\mathscr{E}}(x)$
$\mu_{\mathscr{E}}(\tilde{x})$
$\mathscr{E}$
∇_Θ to (a)
$\max_\Theta \frac{1}{n}\left\|\mu_{\mathscr{E}}(\tilde{x}) - \mu_{\mathscr{E}}(x)\right\|_2^2$
no instruction